PROGRAMING FUNDAMENTAL

LAB NO 06

SUBMITTED FROM: Samreen Zia
SUBMITTED TO: Mr. Rehan Ahmed
DEPARTMENT: Software Engineering (1A)
DATE : 04 Dec 2024

TASK # 01
Run the sample programs, note the output and get familiar with the syntax of for loop.

SAMPLE NO 1:

```
#include "stdafx.h"
#include<iostream>
using namespace std;
int main()
{
cout<<"Use of for loop"<<endl;
int i;
for(i=0;i<10;i++)
{
cout<<i<<endl;
}
system("pause");
return 0;
}
```

OUTPUT:

```
Use of for loop
0
1
2
3
4
5
6
7
8
9
Press any key to continue . . .
```

SAMPLE NO 2:

```
#include "stdafx.h"
#include<iostream>
using namespace std;
int main()
{ cout<<"use of nested loops"<<endl;
 for(int i=1;i<=4;i++)
 {
 for(int j=1;j<=3;j++)
 {
 cout<<i<<"\t"<<j<<endl;
 }
 }
 system("pause");
return 0;
}
```

OUTPUT:

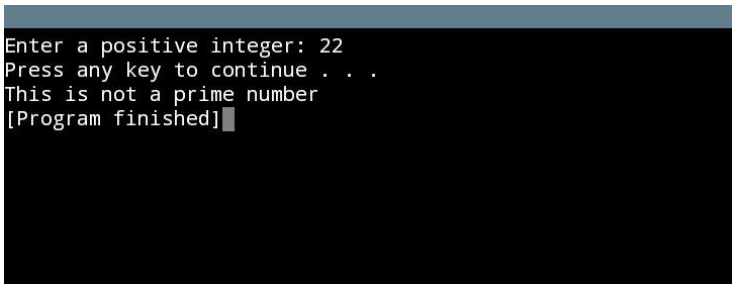
```
use of nested loops
1       1
1       2
1       3
2       1
2       2
2       3
3       1
3       2
3       3
4       1
4       2
4       3
Press any key to continue . . .
```

SAMPLE NO 3:

```
#include "stdafx.h"
#include<iostream>
using namespace std;
```

```
int main()
{
 int n, i, count=0;
 cout << "Enter a positive integer: ";
 cin >> n;
 for(i=2;i<n;i++)
 {
 if(n%i==0)
 { count=1;
break;
 }
 }
 if (count==0)
 cout << "This is a prime number";
 else
 cout << "This is not a prime number";
 system("pause");
return 0;
}
```

OUTPUT:



TASK # 02

Create a program which print the table of a number using for loop .
table of and table upto will be entered by user

SAMPLE OUTPUT

Please Enter table of? 3
Please enter table upto ? 20
3x1=3
3x2=6
……..
3x20=60

PROGRAM:

```
#include<iostream>
```

```
using namespace std;

int main()
{
   int i , n;
   cout<<"Enter a number n ";
   cin>>n;
      for (i=1;i<=20;i++)
   cout<<n<<"*"<<i<<"="<<n*i<<endl;
   return 0;

}
```

OUTPUT:

```
Enter a number n 3
3*1=3
3*2=6
3*3=9
3*4=12
3*5=15
3*6=18
3*7=21
3*8=24
3*9=27
3*10=30
3*11=33
3*12=36
3*13=39
3*14=42
3*15=45
3*16=48
3*17=51
3*18=54
3*19=57
3*20=60

[Program finished]
```

TASK # 03

Create a c++ to find the sum of first 50 natural numbers using for loop

PROGRAM:

```
#include "stdafx.h"
#include<iostream>
using namespace std;
int main()
{

        int sum=0;
        int i ;
        for (i=1; i<=50; i++)
        sum = sum + i;

        cout<<"sum of first 50 natural number = "<<sum ;
```

```
        return 0;
}
```

OUTPUT:

```
sum of first 50 natural number = 1275
[Program finished]
```

TASK # 04

Create a C++ program to print the following * patterns using for loop

Sample Output:

```
*
**
***
****
*****
******
```

PROGRAM:

```
#include<iostream>
using namespace std;

int main(int argc, char *argv[])
{
    int i,j;
   for( i=1 ; i<=6;i++)
{   for (j=1;j<=i ;j++)
   cout<<"*";
  cout<<endl;}
    return 0;
}
```

OUTPUT:

```
*
**
***
****
*****
******

[Program finished]
```

TASK # 05

Create a program which takes 10 values from the user using a for loop. Your program should count the total number of positive numbers, total number of negative numbers and total number of zeros.

PROGRAM:

```
#include <iostream>
using namespace std;

int main()
{
   int num, positive = 0, negative = 0, zero = 0;

   cout << "Enter 10 numbers: " << endl;
   for(int i = 0; i < 10; i++)
   {
      cin >> num;
      if(num > 0)
 {
       positive++;
      } else if(num < 0)
 {
          negative++;
      }
 else {
          zero++;
      }
   }
   cout << "Total positive numbers: " << positive << endl;
   cout << "Total negative numbers: " << negative << endl;
   cout << "Total zeros: " << zero << endl;

   return 0;
}
```

OUTPUT:

```
Enter 10 numbers:
12
-2
-3
4
6
-22
10
4
-7
1
Total positive numbers: 6
Total negative numbers: 4
Total zeros: 0

[Program finished]
```

TASK # 06

Write a program to print abc using their ascii values.
// ascii value for a=97

PROGRAM:

```
#include <iostream>
using namespace std;

int main()
{
    char a = 97;  // ASCII value of 'a'
    char b = 98;  // ASCII value of 'b'
    char c = 99;  // ASCII value of 'c'
    cout << a << b << c << endl;

    return 0;
}
```

OUTPUT:

```
abc

[Program finished]
```

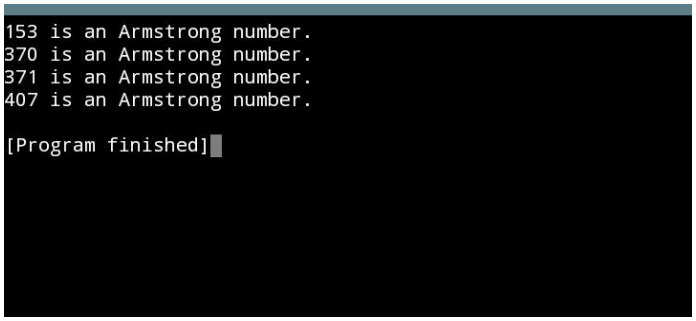
TASK # 07

Write a program to print out all Armstrong numbers between 99 and 999. If sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number**. For example,** 153 = (1 * 1 * 1) + (5 * 5* 5) + (3 * 3 * 3)

PROGRAM:

```
#include <iostream>
using namespace std;

int main()
{
   for(int num = 100; num <= 999; num++)
 {
       int originalNum = num;
       int sum = 0;
       int digit;
       while(originalNum != 0)
 {
           digit = originalNum % 10;
           sum += digit * digit * digit;
           originalNum /= 10;
       }
       if(sum == num)
{
           cout << num << " is an Armstrong number." << endl;
       }
   }

   return 0;
}
```

OUTPUT: